Servicing – General Data

1. General

Safety precautions to be observed by personnel working on the aircraft are given, if required, as part of the individual maintenance procedures. Danger areas are identified on the aircraft by red stencil markings, e.g. BEWARE PROPELLERS, marked on forward face of pilot's and co-pilot's door jamb.

2. Application of DC Power for Ground Servicing

CAUTION DO NOT COMMENCE WORK ON AIRCRAFT ELECTRICAL EQUIPMENT WITH ELECTRICAL POWER CONNECTED, UNLESS WORK CONSISTS OF TESTS REQUIRING USE OF ELECTRICAL POWER.

The following descriptions of external power, ground power and GND PWR all refer to the application of 28 Volt DC power to the aircraft using the external power receptacle. The external power receptacle is located on the left side of the fuselage aft of the cargo door. A spring loaded flap is fitted over the power receptacle to protect it from dirt and weather. For aircraft incorporating S.O.O. 6371, the external ground power receptacle is located forward with the main battery in the outer metallic nose structure, at approximately STA 55. A cover is fitted and is secured with quarter turn fasteners over the power receptacle to protect it from dirt, weather and water ingress.

A. Application of DC Power (Pre Mod 6/2314)

DC Power may be provided to the aircraft using either external power or battery power. If battery power is used, amount of time in use should be minimized to prevent excessive battery discharge.

1) To apply external DC Power proceed as follows:

a) Ensure that POWER SOURCE switch is set to OFF.

b) Connect 28 Volt DC power source to aircraft external DC power receptacle.

c) Ensure circuit breakers are engaged.

d) Set POWER SOURCE switch to EXTERNAL.

e) Set MASTER switch to ON; set BUS TIE switch to NORMAL.

f) After the cockpit display screens power on, check the ELECTRICAL window. GND PWR should be ON (green) and ground power voltage is displayed in the lower left.

g) When ground servicing is complete, disconnect DC Power:

1 Set POWER SOURCE switch to OFF.

2 Set MASTER switch to OFF.

2) To apply DC Power from the battery proceed as follows:

a) Ensure circuit breakers are engaged.

b) Set POWER SOURCE switch to BATTERY.

c) Set MASTER switch to ON; set BUS TIE switch to NORMAL.

d) After the cockpit display screens power on, check the ELECTRICAL window. Battery VOLTS and AMPS are displayed in the lower area of the window. Negative AMPS readings indicate battery discharge (e.g. 34 A).

e) When ground servicing is complete, disconnect DC Power:

1 Set POWER SOURCE switch to OFF.

2 Set MASTER switch to OFF.

B. Application of DC Power (Post Mod 6/2314)

DC Power may be provided to the aircraft using external power, battery power or both. If battery power is used without external power, amount of time in use should be minimized to prevent excessive battery discharge.

1) To apply external DC Power proceed as follows:

a) Ensure that EXTERNAL switch is set to OFF.

b) Ensure that AVIONICS START BATTERY switch is set to OFF (Post Mod 6/2315).

c) Connect 28 Volt DC power source to aircraft external DC power receptacle.

d) Ensure circuit breakers are engaged.

e) Set EXTERNAL switch to ON and confirm that the BUS TIE switch is set to NORMAL.

f) After the cockpit display screens power on, check the ELECTRICAL window. GND PWR should be ON (green) and ground power voltage is displayed in the lower left.

g) When ground servicing is complete, disconnect DC Power:

1 Set EXTERNAL switch to OFF.

2) To apply DC Power from the battery proceed as follows:

a) Ensure circuit breakers are engaged.

b) Ensure that AVIONICS START BATTERY switch is set to OFF (Post Mod 6/2315).

c) Set BATTERY switch to ON and confirm that the BUS TIE switch is set to NORMAL.

d) After the cockpit display screens power on, check the ELECTRICAL window. Battery VOLTS and AMPS are displayed in the lower area of the window. Negative AMPS readings indicate battery discharge (e.g. 34 A).

e) When ground servicing is complete, disconnect DC Power:

1 Set BATTERY switch to OFF.

3) To apply DC Power from the battery and external proceed as follows:

a) Ensure that BATTERY and EXTERNAL switches are both set to OFF.

b) Ensure that AVIONICS START BATTERY switch is set to OFF (Post Mod 6/2315).

c) Connect 28 Volt DC power source to aircraft external DC power receptacle.

d) Ensure circuit breakers are engaged.

e) Set BATTERY switch to ON and confirm that the BUS TIE switch is set to NORMAL.

f) After the cockpit display screens power on, check the ELECTRICAL window. Battery VOLTS and AMPS are displayed in the lower area of the window. If external voltage is present at the DC power receptacle, the bottom of the screen will display GROUND POWER AVAILABLE and the ground power voltage is displayed.

DE HAVILLAND AIRCRAFT OF CANADA LIMITED

TEMPORARY REVISION

PSM: 1-64-2 **AIRCRAFT: DHC-6** **SERIES: 400**

PUBLICATION: MAINTENANCE MANUAL **TEMP. REVISION NO.: TR12-22**

Insert in 12-00-00 after Page 2

SERVICING - GENERAL DATA

REASON FOR ISSUE:

To introduce S.O.O. 6478, Wing Hardpoint Installation, NATO, STA 228.75 and 266.25.

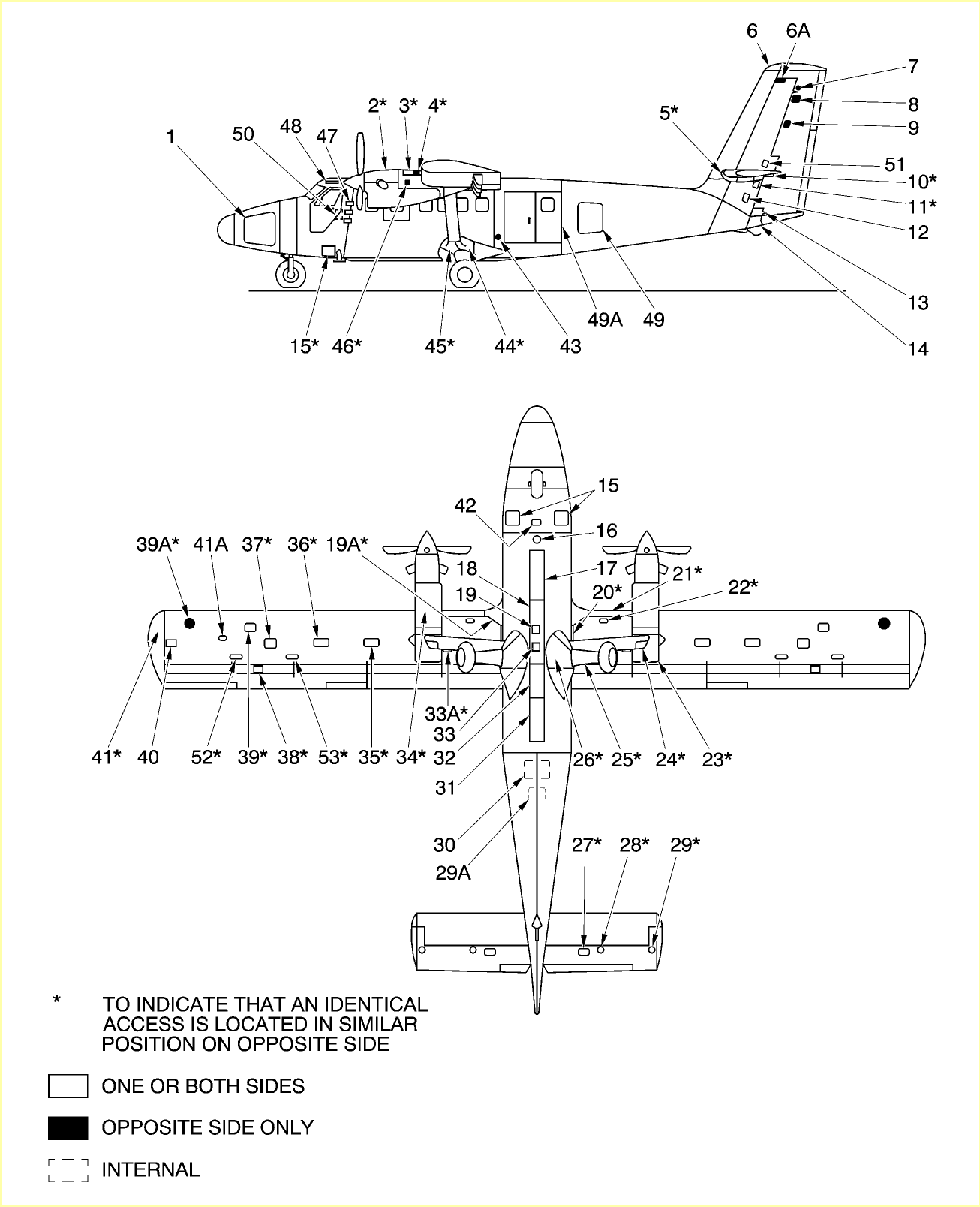
ACTION:

In Chapter 12, Section 12-00-00, ignore existing Pages 4 to 9 and insert revised page as shown on Pages 2 to 8 of this Temporary Revision.

-Continued overleaf-

RECORDING INSTRUCTIONS:

Record this TEMPORARY REVISION NO. and the relevant information in the appropriate columns of the TEMPORARY REVISION INDEX provided at the front of the manual.



Access and Inspection Provisions
Figure 1

Table 1 Key to Figure 1. Access and Inspection Provisions

1.	Nose Gear Shock Strut Upper Mounting Bolts Air Inflation Valve Nose Wheel Steering Actuator, Cable and Pulleys Hydraulic Lines Static Manifold, Pitot and Static Lines Caution Light Control Panel Nose Compartment Utility Light Electrical Wiring Radio Equipment Fwd Avionics Shelves and Maintenance Panel
2.	Propeller Governor, Propeller Overspeed Governor, Breather Propeller Control Propeller Tachometer Generator Engine Front Lifting Eye Vibration Isolators Spark Ignitors Torque Pressure Transmitter
3.	Engine Control Cables and Pulleys Engine Rear Lifting Eye Starter-Generator and Wiring Fuel Pump Oil Scavenge Return Line Oil Tank Filler Cap and Dipstick
4.	Engine Controls
5.	Vertical Stabilizer Forward Attachment Horizontal Stabilizer Forward Attachment Electrical Wiring Apex Magnetometer x2 (Post Mod 6/2171)
6.	Anti-Collision Light
6A.	VOR/LOC Antenna Cable Harness (Mod 6/1473)
7.	Rudder Top Hinge
8.	Rudder Trim Tab Screw Jack
9.	Rudder Trim Tab Turnbuckles
10.	Horizontal Stabilizer Rear Attachment Elevator and Flap/Elevator Interconnect Trim Cables Elevator Torque Tube Bellcrank and Support Bracket
11.	Elevator Quadrant Rudder Trim Tab Control Cables
12.	Rudder Cables and Pulleys Elevator Trim Tab Control Cables and Pulleys

Table 1 Key to Figure 1. Access and Inspection Provisions (continued)

13. Rudder Control Cable Attachment Bracket
Rudder Lower Hinge
Rudder Stops
Rudder Geared Tab Gearbox

14. Tail Bumper Shock Strut Attachment
Tail Light Electrical Wiring

15. Hydraulic Power Package
Flying Control Cables and Pulleys
Rudder Gust Lock Mechanism
Electrical Wiring
Heating System:

- Hot Air Valve
- Expansion Chamber
- Recirculated Air Duct
- Silencer
- Ram Air Valve
- Cabin Air Control Valve
- Ejector
- Main Duct Fan

16. Electrical Wiring

17. No. 1, 2, and 3 Fuel Cells, Sump Plates and Tank Units
Forward Tank Interconnector Manifold
Forward Tank Fuel Ejector

18. No. 4, 5, and 6 Fuel Cells, Sump Plates and Tank Units
Forward and Aft Tank Interconnect Drain Valves
Forward and Aft Tank Booster Pumps
Fuel Pressure Switches
Crossfeed Valve

19. Forward Tank Drain Valve
Forward Tank Booster Pumps Drain Valves

19A. Wing-to-Fuselage Electrical Connectors

20. Wing-to-Fuselage Front and Rear Attachments
Aileron Cables
Bleed Air Line
Flap Push-Pull Rod
Electrical Wiring
De-icing Line

21. Engine Control Cables and Fairleads
Electrical Wiring

22. Bleed Air Line
De-icing Line

Table 1 Key to Figure 1. Access and Inspection Provisions (continued)

23.	Wing Strut Upper Attachment Fuel Main Feed Line
24.	Wing Strut
25.	Brake Hydraulic Line Wheel-Ski Hydraulic Lines and Electrical Wiring (if fitted)
26.	Main Landing Gear Attachment Shock Absorber Brake Hydraulic Line Jacking Point
27.	Flap/Elevator Interconnect Jack (Right Side) Elevator Trim Jack (Left Side)
28.	Elevator Mid Hinge Attachment
29.	Elevator Outboard Hinge Attachment
29A.	Rudder and Elevator Cables
30.	Main Battery
31.	No. 8 Fuel Cell. Sump Plate, and Tank Unit Aft Tank Fuel Ejector Aft Tank Interconnector Manifold
32.	No. 6 and 7 Fuel Cells, Sump Plate, and Tank Unit Aft Tank Interconnect Manifold
33.	Aft Tank Drain Valve Aft Tank Booster Pumps Drain Valves
33A.	Aileron Control Cables and Fairleads Flap Inboard Operating Bellcrank
34.	Fuel Strainer and Fuel Lines Fuel Purge Connections Firewall Shut-off Valve Fire Extinguisher Bottle Ignition Current Regulator Box Gas Generator Tachometer Generator Oil Cooler and Oil Lines Vibration Isolators Electrical Wiring
35.	Aileron Control Cables and Fairleads Flap Idler Lever and Push-Pull Rod
36.	Aileron Control Cables and Fairleads Flap Idler Lever and Push-Pull Rod

Table 1 Key to Figure 1. Access and Inspection Provisions (continued)

37. Aileron Control Cables and Pulleys
Flap Outboard Operating Bellcrank

38. Aileron Linkage and Flap Outboard Attachment

39. Wing Electrical Wiring Conduit
De-icing Distributor Valve
Wing De-icing Water Separator (Mod 6/1440)

39A. Wing Tank Components (Mod S.O.O. 6247)

40. Wing Tank Capacitance Unit Level Control (Mod S.O.O. 6247)

41. Navigation Light
Wing Tank Components (Mod S.O.O. 6247)

- Wing Tank Low Pressure Switch
- Fuel Transfer Valve
- Check Valve
- Pressure Pump
- Refuel Shut-off Valve
- Radio Noise Filter (Mod 6/1374)
- Wing Tank Fuel Filter (Mod 6/2006)

41A. Standby Magnetometer One Side (Mod 6/2049) Apex Magnetometer x2 other side (Mod 6/2049) (Pre Mod 6/2171)

42. Hydraulic System Filter

43. Rudder and Elevator Control Cables and Pulleys

44. Fuel Main Feed Line
Wheel Brake Hydraulic Line
Shock Absorber

45. Wing Strut Lower Attachment

46. Engine Oil Filter

47. Roof Access Steps

48. Handle Hold

49. Aft Avionics Shelves and Maintenance Panel at Station 421.00
Rudder and Elevator Servo's (S.O.O. 6221)
Rudder Trim Servo (S.O.O. 6221)

49A. For Aircraft with S.O.O. 6393:
Aft Avionics Shelves and Maintenance Panel at Station 421.00
Rudder and Elevator Servo's (S.O.O. 6221)
Rudder Trim Servo (S.O.O. 6221)

50. Aileron Servo (Co-pilot side) (S.O.O. 6221)

Table 1 Key to Figure 1. Access and Inspection Provisions (continued)

	51.	Elevator Position Transmitter (if fitted with Mod 6/2417)
▌	52.	NATO Hardpoints, wiring access panel at Station 266.25 (S.O.O. 6478)
▌	53.	NATO Hardpoints, wiring access panel at Station 228.75 (S.O.O. 6478)

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g) If the ground power voltage is higher than the battery voltage (between approximately one and five volts higher), then the EXTERNAL may be switched ON.

h) Review the ELECTRICAL display screen. GND PWR should be green, BATTERY VOLTS and GND PWR VOLTS should be approximately equal (within 0.2 volts) and BATTERY AMPS should be a positive number, indicating that the battery is being charged by the GND PWR.

Note Avoid excessive battery charge currents using external power. Typically charging currents should be below 50 AMPS within one or two minutes of BATTERY and EXTERNAL both being switched ON.

i) When ground servicing is complete, disconnect DC Power:

1 Set BATTERY switch to OFF.

2 Set EXTERNAL switch to OFF.

3. Access and Inspection Provisions

CAUTION BEFORE REFITTING ACCESS PANELS, CHECK FOR INTERNAL CLEANLINESS TO ENSURE THAT NO TOOLS, LOCKWIRE, ETC. HAVE BEEN LEFT INSIDE THE STRUCTURE. ALSO CHECK THAT FUEL DRAIN VALVES ARE LOCKWIRED IN THE CLOSED POSITION.

Removable panels necessary for access to equipment for inspection and maintenance purposes are shown in Figure 1 and Table 1 — Key to Figure 1. Access and Inspection Provisions. Interior upholstery is removable in all places where access is necessary for maintenance.

4. Servicing Provisions

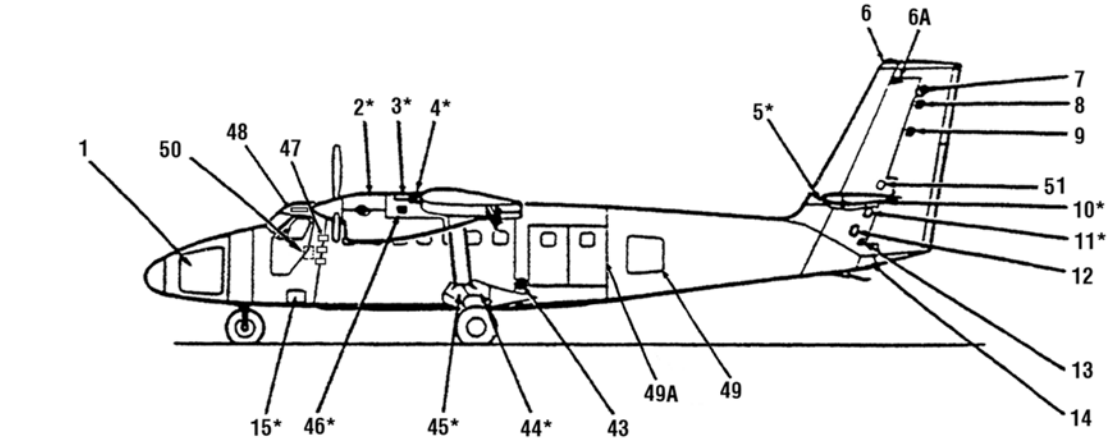
For servicing provisions refer to Figure 2.

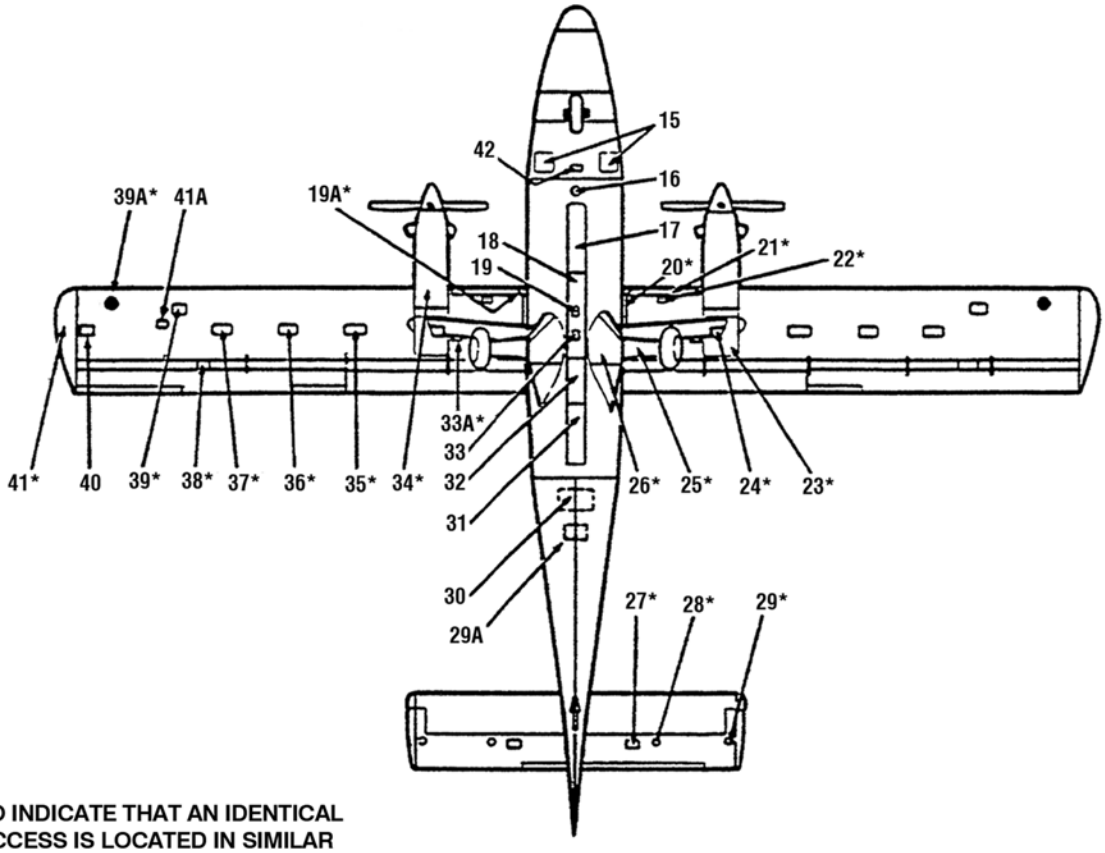
5. Application of Roof Access Steps

The roof access steps are located on the left side of the fuselage aft of the cockpit door. A handle hold is installed on the roof panel above the cockpit door to aid the usage of the roof access steps.

The function of the roof access steps is to provide easy access to the upper fuselage panels.

WARNING PROPER FALL ARREST SAFETY EQUIPMENT SHOULD BE USED WHEN ACCESS IS REQUIRED TO THE UPPER FUSELAGE.





* TO INDICATE THAT AN IDENTICAL ACCESS IS LOCATED IN SIMILAR POSITION ON OPPOSITE SIDE.

ONE OR BOTH SIDES

OPPOSITE SIDE ONLY

INTERNAL

Access and Inspection Provisions
Figure 1

Table 1 Key to Figure 1. Access and Inspection Provisions

1.	Nose Gear Shock Strut Upper Mounting Bolts Air Inflation Valve Nose Wheel Steering Actuator, Cable and Pulleys Hydraulic Lines Static Manifold, Pitot and Static Lines Caution Light Control Panel Nose Compartment Utility Light Electrical Wiring Radio Equipment Fwd Avionics Shelves and Maintenance Panel
2.	Propeller Governor, Propeller Overspeed Governor, Breather Propeller Control Propeller Tachometer Generator Engine Front Lifting Eye Vibration Isolators Spark Ignitors Torque Pressure Transmitter
3.	Engine Control Cables and Pulleys Engine Rear Lifting Eye Starter-Generator and Wiring Fuel Pump Oil Scavenge Return Line Oil Tank Filler Cap and Dipstick
4.	Engine Controls
5.	Vertical Stabilizer Forward Attachment Horizontal Stabilizer Forward Attachment Electrical Wiring Apex Magnetometer x2 (Post Mod 6/2171)
6.	Anti-Collision Light
6A.	VOR/LOC Antenna Cable Harness (Mod 6/1473)
7.	Rudder Top Hinge
8.	Rudder Trim Tab Screw Jack
9.	Rudder Trim Tab Turnbuckles

Table 1 Key to Figure 1. Access and Inspection Provisions (continued)

10. Horizontal Stabilizer Rear Attachment
Elevator and Flap/Elevator Interconnect Trim Cables
Elevator Torque Tube Bellcrank and Support Bracket

11. Elevator Quadrant
Rudder Trim Tab Control Cables

12. Rudder Cables and Pulleys
Elevator Trim Tab Control Cables and Pulleys

13. Rudder Control Cable Attachment Bracket
Rudder Lower Hinge
Rudder Stops
Rudder Geared Tab Gearbox

14. Tail Bumper Shock Strut Attachment
Tail Light Electrical Wiring

15. Hydraulic Power Package
Flying Control Cables and Pulleys
Rudder Gust Lock Mechanism
Electrical Wiring
Heating System:

- Hot Air Valve
- Expansion Chamber
- Recirculated Air Duct
- Silencer
- Ram Air Valve
- Cabin Air Control Valve
- Ejector
- Main Duct Fan

16. Electrical Wiring

17. No. 1, 2, and 3 Fuel Cells, Sump Plates and Tank Units
Forward Tank Interconnector Manifold
Forward Tank Fuel Ejector

18. No. 4, 5, and 6 Fuel Cells, Sump Plates and Tank Units
Forward and Aft Tank Interconnect Drain Valves
Forward and Aft Tank Booster Pumps
Fuel Pressure Switches
Crossfeed Valve

Table 1 Key to Figure 1. Access and Inspection Provisions (continued)

19.	Forward Tank Drain Valve Forward Tank Booster Pumps Drain Valves
19A.	Wing-to-Fuselage Electrical Connectors
20.	Wing-to-Fuselage Front and Rear Attachments Aileron Cables Bleed Air Line Flap Push-Pull Rod Electrical Wiring De-icing Line
21.	Engine Control Cables and Fairleads Electrical Wiring
22.	Bleed Air Line De-icing Line
23.	Wing Strut Upper Attachment Fuel Main Feed Line
24.	Wing Strut
25.	Brake Hydraulic Line Wheel-Ski Hydraulic Lines and Electrical Wiring (if fitted)
26.	Main Landing Gear Attachment Shock Absorber Brake Hydraulic Line Jacking Point
27.	Flap/Elevator Interconnect Jack (Right Side) Elevator Trim Jack (Left Side)
28.	Elevator Mid Hinge Attachment
29.	Elevator Outboard Hinge Attachment
29A.	Rudder and Elevator Cables
30.	Main Battery

Table 1 Key to Figure 1. Access and Inspection Provisions (continued)

31.	No. 8 Fuel Cell. Sump Plate, and Tank Unit Aft Tank Fuel Ejector Aft Tank Interconnector Manifold
32.	No. 6 and 7 Fuel Cells, Sump Plate, and Tank Unit Aft Tank Interconnect Manifold
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36.	Aileron Control Cables and Fairleads Flap Idler Lever and Push-Pull Rod
37.	Aileron Control Cables and Pulleys Flap Outboard Operating Bellcrank
38.	Aileron Linkage and Flap Outboard Attachment
39.	Wing Electrical Wiring Conduit De-icing Distributor Valve Wing De-icing Water Separator (Mod 6/1440)
39A.	Wing Tank Components (Mod S.O.O. 6247)
40.	Wing Tank Capacitance Unit Level Control (Mod S.O.O. 6247)

Table 1 Key to Figure 1. Access and Inspection Provisions (continued)

41. Navigation Light
Wing Tank Components (Mod S.O.O. 6247)

- Wing Tank Low Pressure Switch
- Fuel Transfer Valve
- Check Valve
- Pressure Pump
- Refuel Shut-off Valve
- Radio Noise Filter (Mod 6/1374)
- Wing Tank Fuel Filter (Mod 6/2006)

41A. Standby Magnetometer One Side (Mod 6/2049) Apex Magnetometer x2 other side (Mod 6/2049) (Pre Mod 6/2171)

42. Hydraulic System Filter

43. Rudder and Elevator Control Cables and Pulleys

44. Fuel Main Feed Line
Wheel Brake Hydraulic Line
Shock Absorber

45. Wing Strut Lower Attachment

46. Engine Oil Filter

47. Roof Access Steps

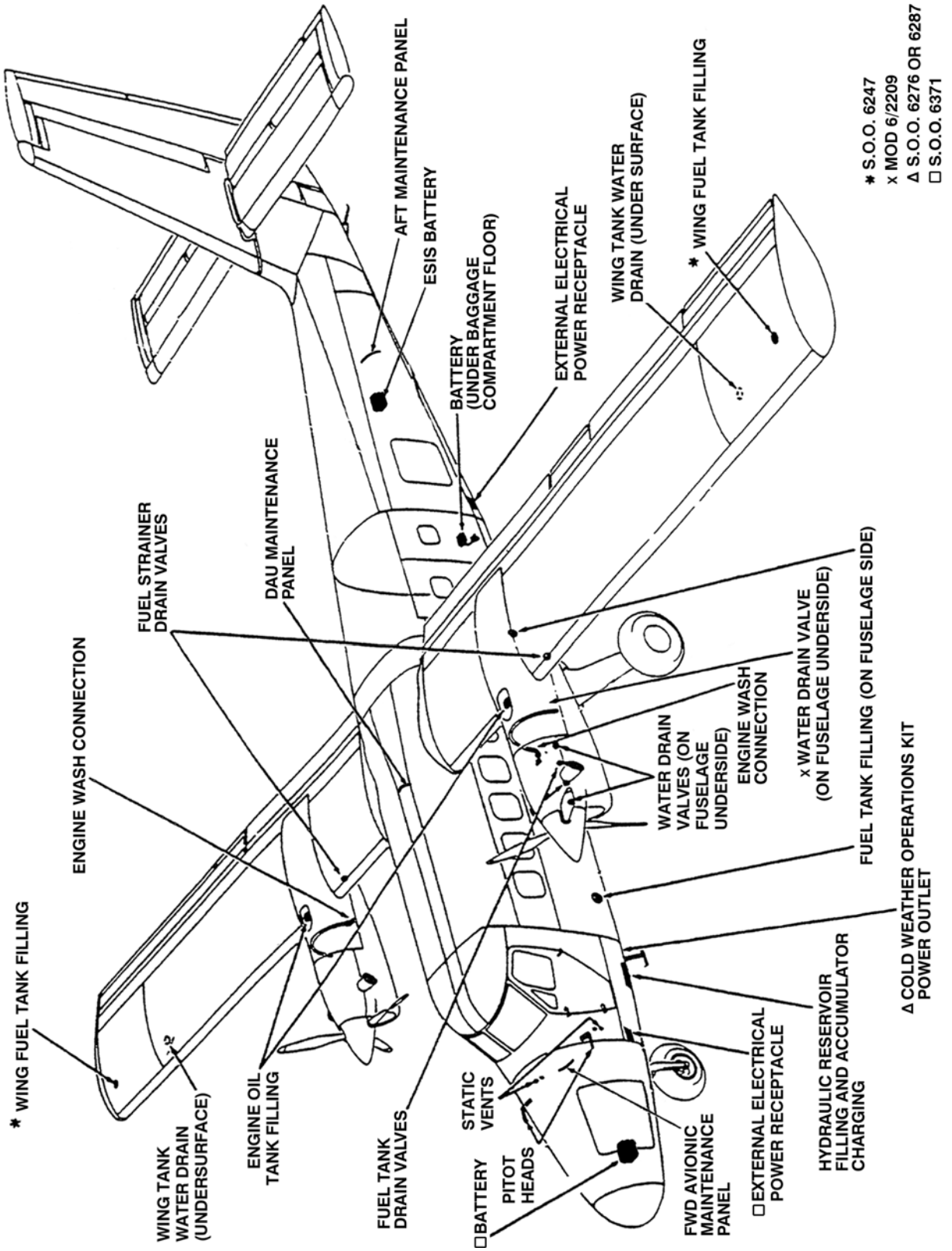
48. Handle Hold

49. Aft Avionics Shelves and Maintenance Panel at Station 421.00
Rudder and Elevator Servo's (S.O.O. 6221)
Rudder Trim Servo (S.O.O. 6221)

49A. For Aircraft with S.O.O. 6393:
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Rudder Trim Servo (S.O.O. 6221)

50. Aileron Servo (Co-pilot side) (S.O.O. 6221)

51. Elevator Position Transmitter (if fitted with Mod 6/2417)



Servicing Diagram
Figure 2

Print Date: 2025-09-12